



Section 5

DC Motor

6. Self-excited DC motor

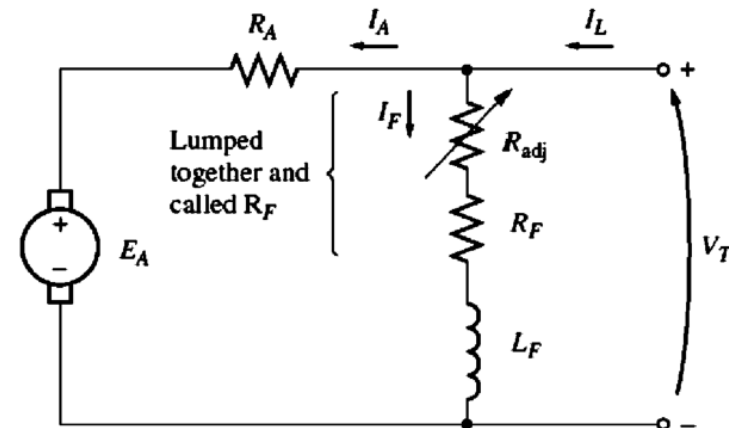
Speed Control of Shunt DC Motors

The two common ways in which the speed of a shunt dc machine can be controlled are:

1. Adjusting the field **resistance** R_F
2. Adjusting the **terminal voltage** applied to the armature

Adjusting the field resistance R_F

- Increasing R_F causes $I_F (= V_T / R_F \uparrow)$ to decrease
- Decreasing I_F decreases ϕ
- Decreasing ϕ lowers $E_A (= K\phi \downarrow \omega)$
- Decreasing E_A increases $I_A (= (V_T - E_A \downarrow) / R_A)$.
- Increasing I_A increases $\tau_{ind} (= K\phi \downarrow I_A \uparrow \uparrow)$, (with the change in I_A dominant over the change in flux)
- Increasing τ_{ind} makes $\tau_{ind} > \tau_{load}$, and the speed ω increases
- Increasing $E_A = K\phi\omega \uparrow$ again
- Increasing E_A decreases I_A .
- Decreasing I_A decreases τ_{ind} until $\tau_{ind} = \tau_{load}$ at a higher speed ω .

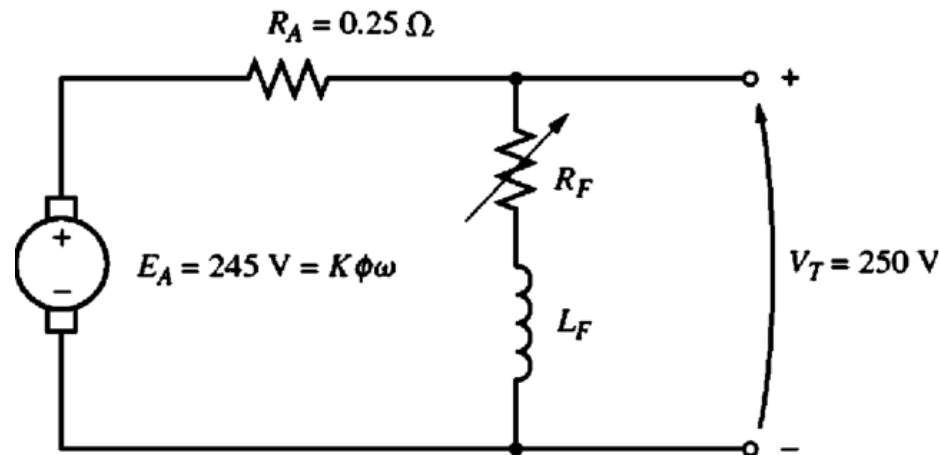


6. Self-excited DC motor

Speed Control of Shunt DC Motors

Adjusting the field resistance R_F (Justify)

A shunt dc motor with an internal resistance of 0.25Ω . It is currently operating with a terminal voltage of 250V and an internal generated voltage of 245V. Therefore, the armature current flow is $I_A = \frac{(250\text{V} - 245\text{V})}{0.25 \Omega} = 20\text{A}$



What happens in this motor if there is a 1 percent decrease in flux?

If the flux decreases by 1 percent, then E_A must decrease by 1 percent too, because $E_A = K\phi\omega$. Therefore, E_A will drop to:

$$E_{A2} = 0.99 E_{A1} = 0.99(245 \text{ V}) = 242.55 \text{ V}$$

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Speed Control of Shunt DC Motors

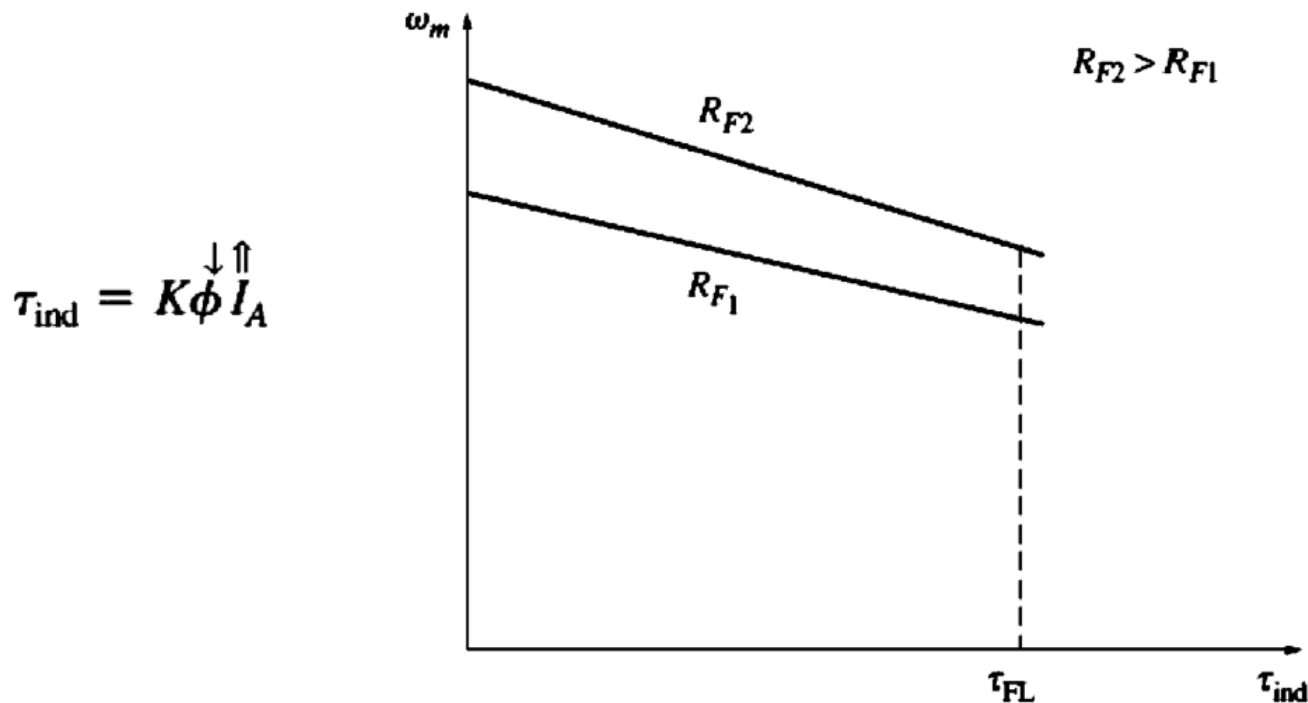
Adjusting the field resistance R_F (Justify)

$$E_{A2} = 0.99 E_{A1} = 0.99(245 \text{ V}) = 242.55 \text{ V}$$

The armature current must then rise to

$$I_A = \frac{250 \text{ V} - 242.55 \text{ V}}{0.25 \Omega} = 29.8 \text{ A}$$

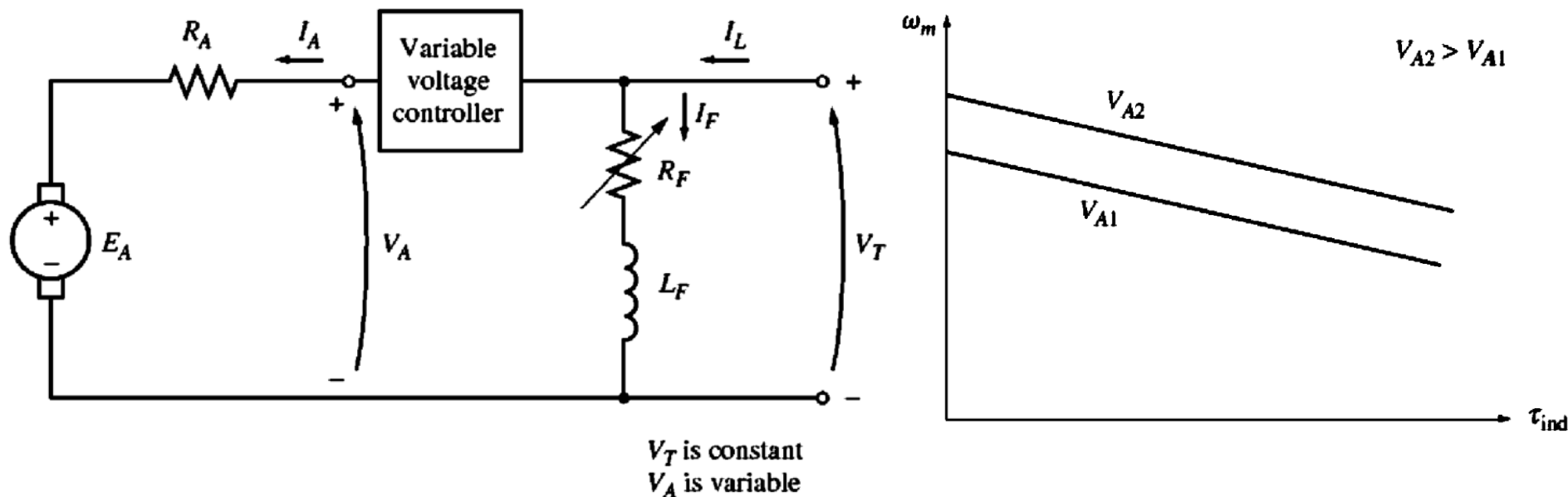
Thus, a **1 percent** decrease in flux produced a **49 percent** increase in armature current.



6. Self-excited DC motor

Speed Control of Shunt DC Motors

Adjusting the terminal voltage applied to the armature



- An increase in V_A increases $I_A = \frac{(V_A \uparrow - E_A)}{R_A}$
- Increasing I_A increases $\tau_{ind} = K\phi I_A \uparrow$
- Increasing τ_{ind} makes $\tau_{ind} > \tau_{load}$, increasing ω
- Increasing ω increases $E_A = K\phi\omega \uparrow$.
- Increasing E_A decreases $I_A = \frac{(V_A - E_A \uparrow)}{R_A}$
- Decreasing I_A decreases τ_{ind} until $\tau_{ind} = \tau_{load}$ at a higher ω .

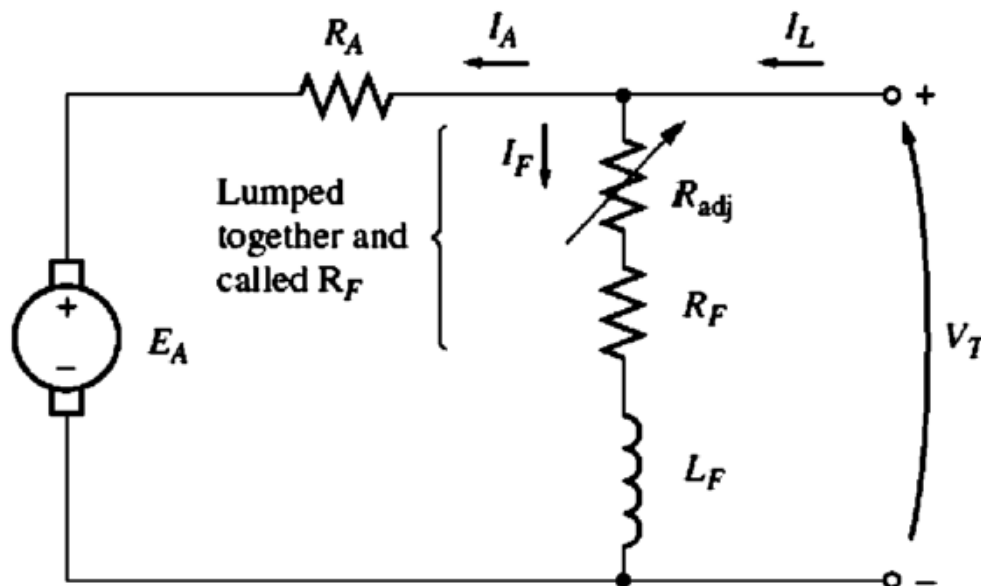
Example 2

A 220-V shunt DC motor has the following parameters:

- Armature resistance: $R_a = 0.5\Omega$
- Field resistance: $R_f = 220\Omega$
- Rated supply voltage: 220 V

The motor runs at 1200rpm and draws an armature current of 20A at rated voltage. Assume $K\phi$ is not change.

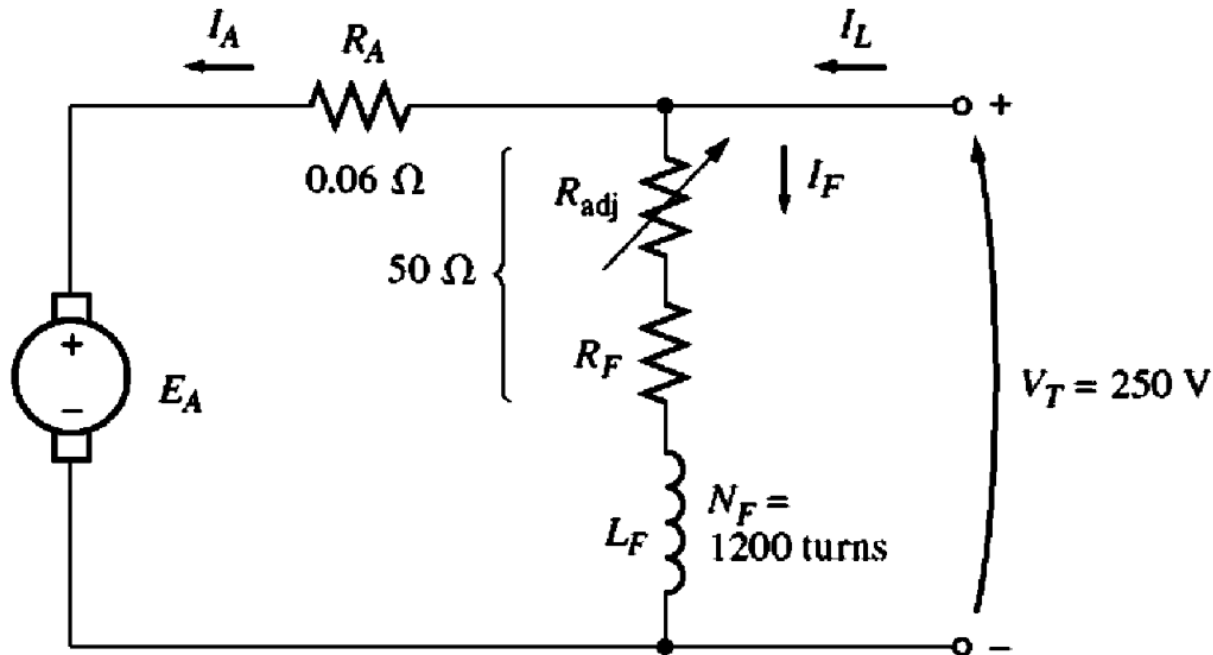
- Calculate the E_A at rated conditions.
- If the supply voltage is reduced to 180V, and the armature current remains 20A, calculate the new speed of the motor.



Example 3

A 50-hp, 250-V, 1200 r/min dc shunt motor with compensating windings has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06Ω . Its field circuit has a total resistance $R_{\text{adj}} + R_F$ of 50Ω , which produces a no-load speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding:

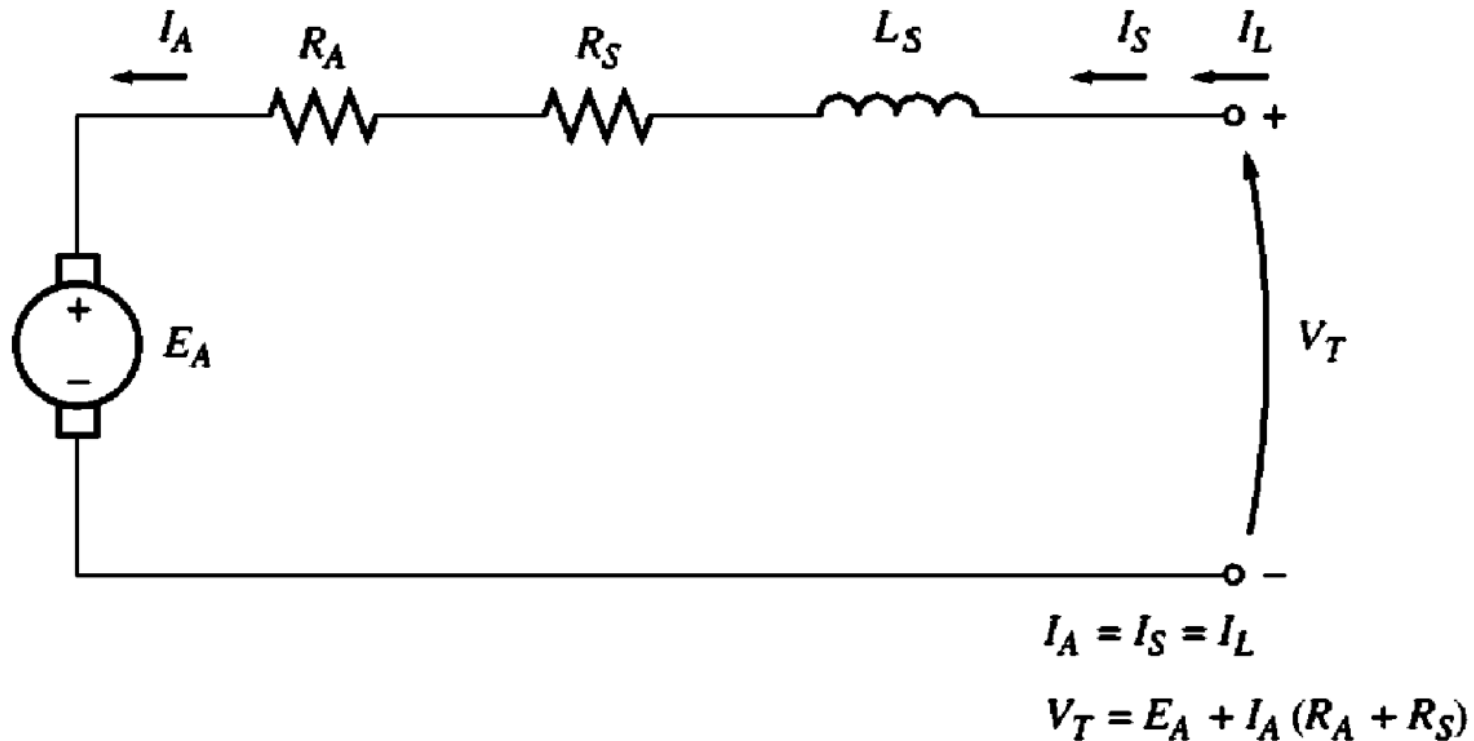
- Find the speed of this motor when its input current is 100 A.
- Find the speed of this motor when its input current is 200 A.
- Find the speed of this motor when its input current is 300 A.
- Plot the torque-speed characteristic of this motor.



6. Self-excited DC motor

Series DC motor

- A series dc motor is a dc motor whose field windings connected in series with the armature circuit.
- The equivalent circuit of a series dc motor is shown:



6. Self-excited DC motor

Series DC motor

The terminal characteristic of a Series DC Motor

- Flux in the motor: $\phi = cI_A$
- Voltage: $V_T = E_A + I_A(R_A + R_S)$
- Induce torque: $\tau_{\text{ind}} = K\phi I_A = KcI_A^2$

$$I_A = \sqrt{\frac{\tau_{\text{ind}}}{Kc}}$$

$$V_T = K\phi\omega + \sqrt{\frac{\tau_{\text{ind}}}{Kc}}(R_A + R_S)$$

- Speed:

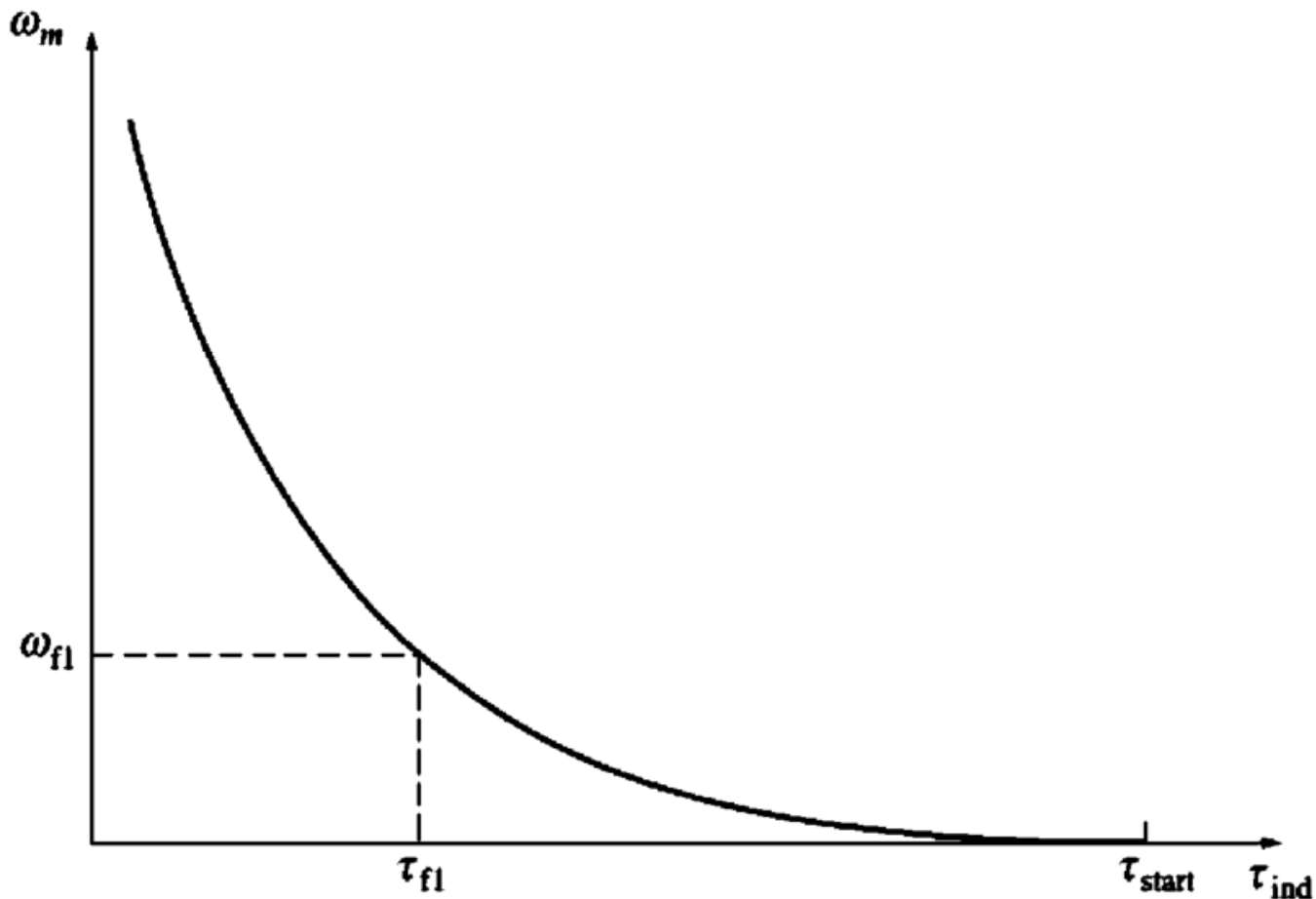
$$\omega = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{\text{ind}}}} - \frac{R_A + R_S}{Kc}$$

6. Self-excited DC motor

Series DC motor

The terminal characteristic of a Series DC Motor

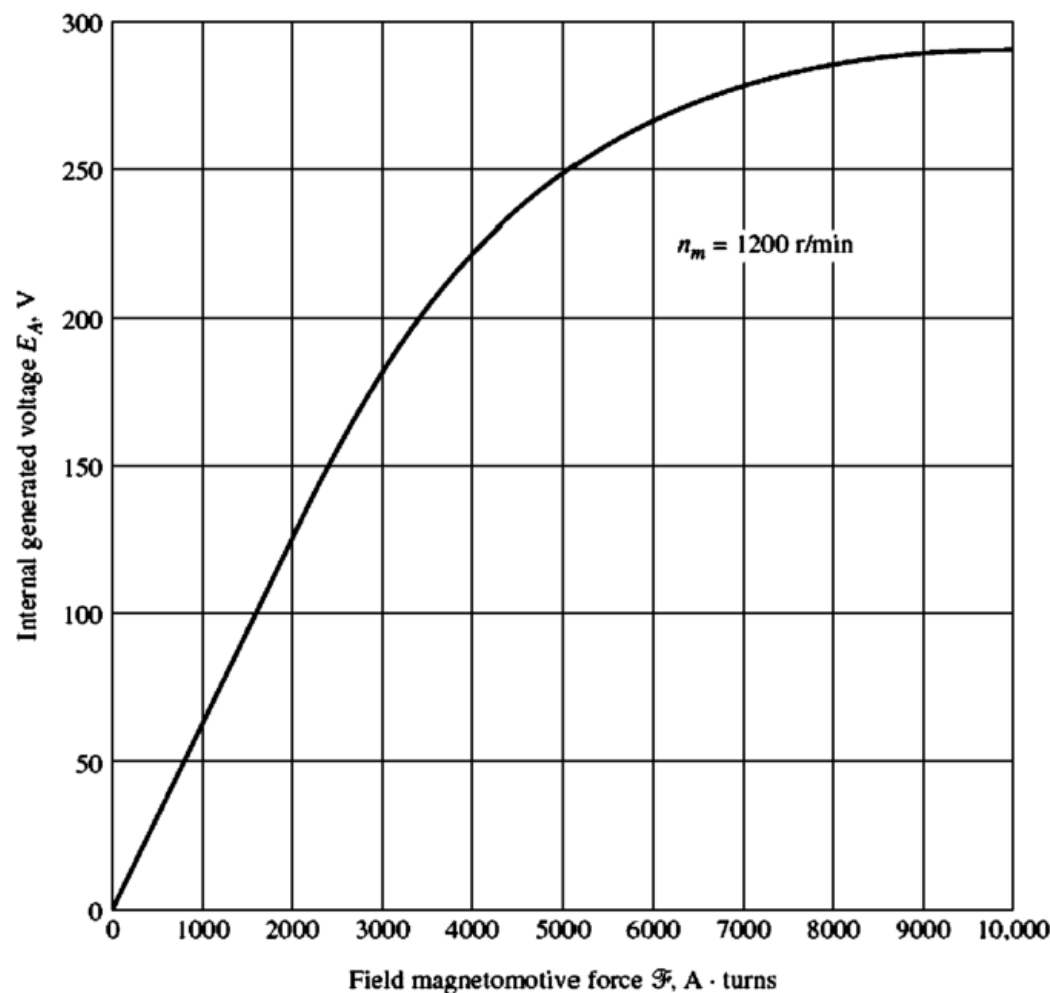
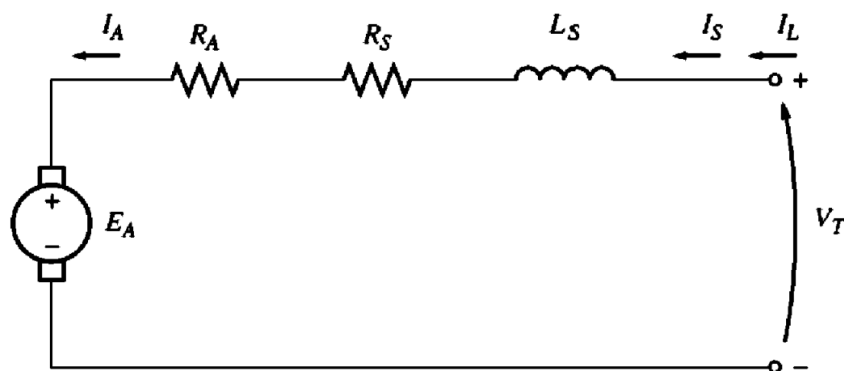
$$\omega = \frac{V_T}{\sqrt{Kc}} \frac{1}{\sqrt{\tau_{\text{ind}}}} - \frac{R_A + R_S}{Kc}$$



Example 4

Figure below shows a 250-V series DC motor with compensating windings, and a total series resistance $R_A + R_S$ of $0.08\ \Omega$. The series field consists of 25 turns per pole, with the magnetization curve is provided. When its armature current is 50 A:

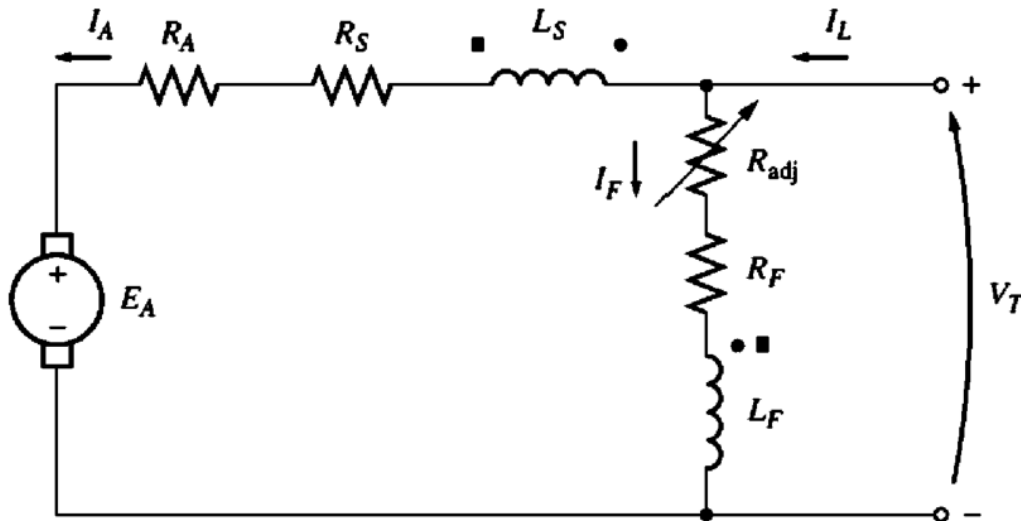
- Find the speed of this motor.
- Determine induced torque of this motor.



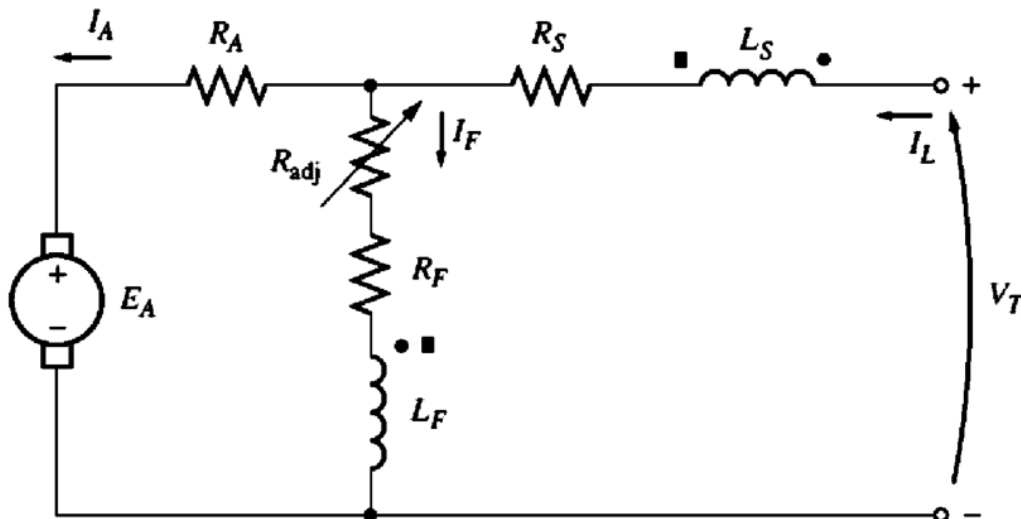
6. Self-excited DC motor

The compounded DC motor

A compounded DC motor is a motor with both a shunt and a series field.



(a)



(b)

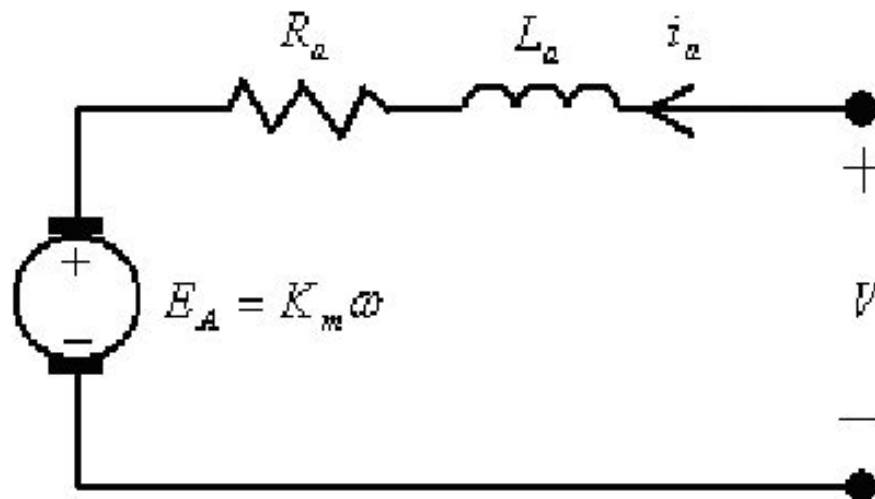
The equivalent circuit of compounded DC motors:

(a) long-shunt connection

(b) short-shunt connection.

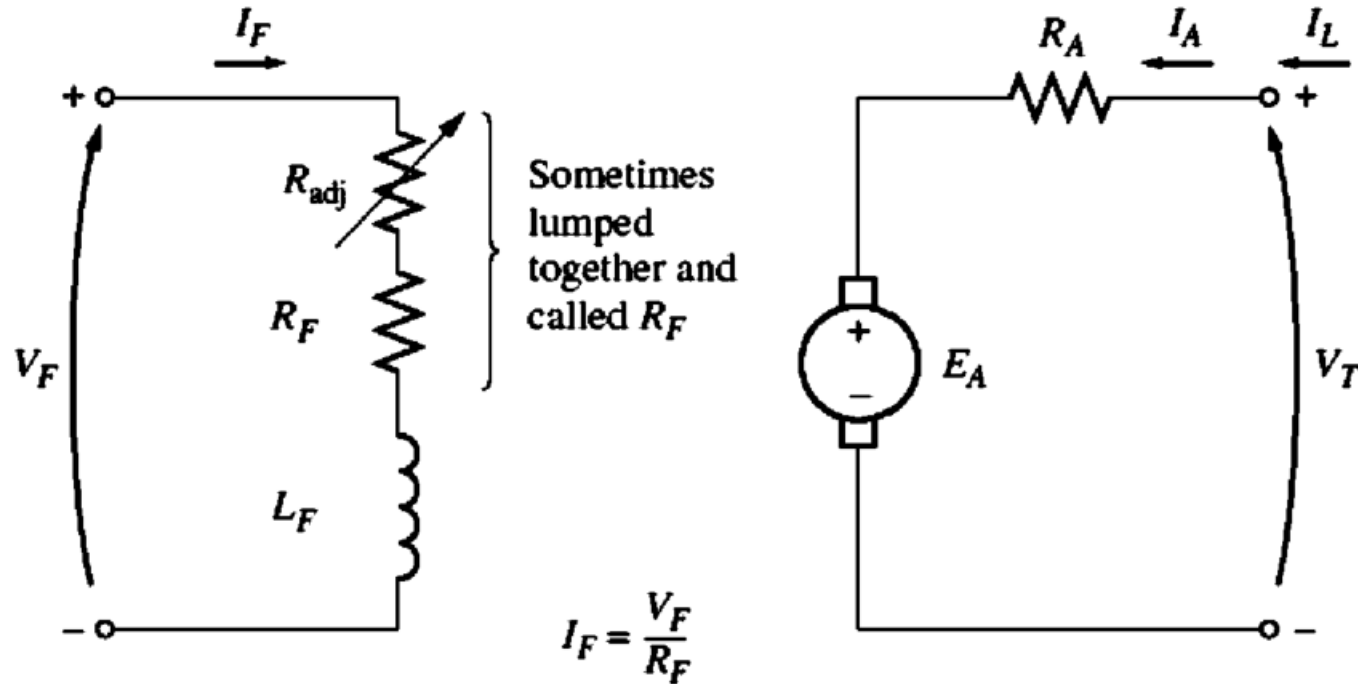
7. The permanent-magnet DC motor

- A permanent-magnet DC (PMDC) motor is a dc motor whose poles are made of permanent magnets.
- These motors do not require an external field circuit, they do not have the field circuit copper losses associated with shunt dc motors.



8. Separately Excited DC motor

- A separately excited dc motor is a motor whose field circuit is supplied from a separate constant-voltage power supply.



$$I_F = \frac{V_F}{R_F}$$

$$V_T = E_A + I_A R_A$$

$$I_L = I_A$$

$$V_T = E_A + I_A R_A$$

Example 5

A separately excited DC motor has the following data:

- Armature voltage, $V_a = 220 \text{ V}$
- Armature resistance, $R_a = 0.5 \Omega$
- Field current is constant (flux is constant)
- Armature current at load, $I_a = 20 \text{ A}$
- Speed at no-load, $n = 1000 \text{ rpm}$
- Armature current at no-load is negligible

Determine:

- a. Internal generated voltage E_A at load.
- b. Speed of the motor at load.
- c. Developed torque at load.